

Srinivasa Manikanta Rajapantula

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SUMMARY

Electrical and Electronics Engineering student with hands-on experience in embedded systems, industrial electrical systems, and IoT. Passionate about developing practical hardware solutions using microcontrollers, sensors, and communication protocols. Seeking opportunities in Embedded Systems, Industrial Automation, and IoT Engineering.

SKILLS

Embedded Systems & Hardware: ESP32, Arduino, Communication Protocols (UART, I2C, SPI, MQTT), Sensors & Actuators

Internet of Things: IoT System Design, Cloud Integration, Real-time Monitoring

Programming Languages: C, Embedded C, Assembly, Python, MATLAB

Power Systems: Power Generation, Transmission & Distribution, Smart Grid Concepts, MATLAB Simulation

Power Electronics: Converters, Inverters, Motor Drives, Thyristors, Choppers, PWM Control

Electrical Machines: DC & AC Machines, Transformers, Electrical Motors

Additional Skills: Github, Firebase, Git

PROJECTS

Firefighter Safety Monitoring System (ESP32 • Firebase • IoT)

- Developed an IoT wearable for real-time firefighter monitoring with environmental sensing, GPS tracking, and cloud dashboard.

Automatic Exhaust Fan (Arduino • MQ-2 • Relay)

- Designed an automated exhaust fan using gas sensing with automatic activation and manual override.

Color Ohm (React Native • Expo)

- Developed a mobile application for resistor color code calculations and electronics reference tools.

Single Axis Solar Tracker (LDR • Relay Logic • DC Gear Motor)

- Designed a relay-based solar tracker improving solar energy capture by approximately 30%.

EXPERIENCE

Industrial Trainee — Electrical & Electronics

Coromandel International Limited, Visakhapatnam | 6th Semester, Diploma — 2024

- Studied 132kV/11kV substation operations including power distribution across multiple plant sections
- Studied Turbo Generator (TG) station — 5MW power generation from waste heat of sulphuric acid plant.

EDUCATION

B.Tech — Electrical & Electronics Engineering | CGPA: 7.5

Andhra Loyola Institute of Engineering & Technology, Vijayawada | 2024 – 2027 (Pursuing)

Diploma — Electrical & Electronics Engineering | 75%

Government Polytechnic Srikakulam | 2021 – 2024

SSC (Class 10) | 542/600 (90.3%)

ZPHS Murapaka, AP State Board | 2021

CERTIFICATIONS

- Drone Technology
- Introduction to Internet of Things (IoT) — NPTEL
- 3D Printing
- Electric Vehicle Technology
- Industrial Internet of Things (IIoT)
- Solar PV System Design